

Array

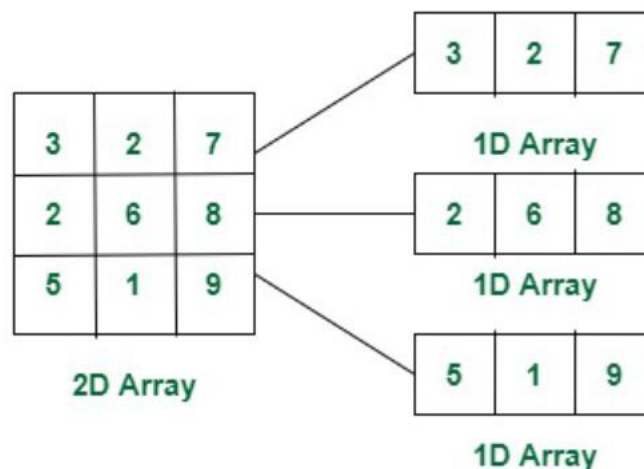
An **array** in Java is a collection of **multiple values of the same data type** stored under a **single variable name**.

One-Dimensional Arrays

```
public class Main {  
    public static void main(String[] args) {  
        int[] arr = {10, 20, 30, 40, 50};  
        System.out.println("Array Elements:");  
        for (int i = 0; i < arr.length; i++) {  
            System.out.println(arr[i]);  
        }  
    }  
}
```

Declaration	Valid?
int[] a;	✓ Recommended
int a[];	✓
int[] a, b;	✓ Both arrays
int a[], b[];	✓ Both arrays
int a[], b;	✓ a array, b int
int a[10];	❑ Invalid in Java

Multidimensional Arrays (2D Array)



```
import java.util.Scanner;  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int[][] a = new int[3][3];  
        System.out.println("Enter 9 elements:");  
        for (int i = 0; i < 3; i++) {  
            for (int j = 0; j < 3; j++) {  
                a[i][j] = sc.nextInt();  
            }  
        }  
        System.out.println("\nMatrix:");  
        for (int i = 0; i < 3; i++) {
```

```

        for (int j = 0; j < 3; j++) {
            System.out.print(a[i][j] + " ");
        }
        System.out.println();
    }
    sc.close();
}

```

Input

10 20 30

40 50 60

70 80 90

Output

Matrix:

10 20 30

40 50 60

70 80 90

Declaration	Valid?
int[][] a;	✓ Recommended
int a[][];	✓
int[] a[];	✓
int[][] a, b;	✓ Both 2D arrays
int[] a[], b;	✓ a 2D array, b 1D array
int[][] a = new int[3][4];	✓
int a[][] = new int[3][4];	✓

Jagged Array in Java

A Jagged Array (also called a **Ragged Array**) is a 2D array in which each row can have a different number of columns.

- Creating odd number of columns in a 2D array, it is known as a jagged array.
- In other words, it is an array of arrays with **different number of columns**.



```

public class Main {

    public static void main(String[] args) {

        int[][] a = {
            {10, 20, 30},
            {40, 50, 60, 70, 80},
            {90, 100}
        };

        for (int i = 0; i < a.length; i++) { //no of rows

            for (int j = 0; j < a[i].length; j++) { //no of elements in the particular row
                System.out.print(a[i][j] + " ");
            }

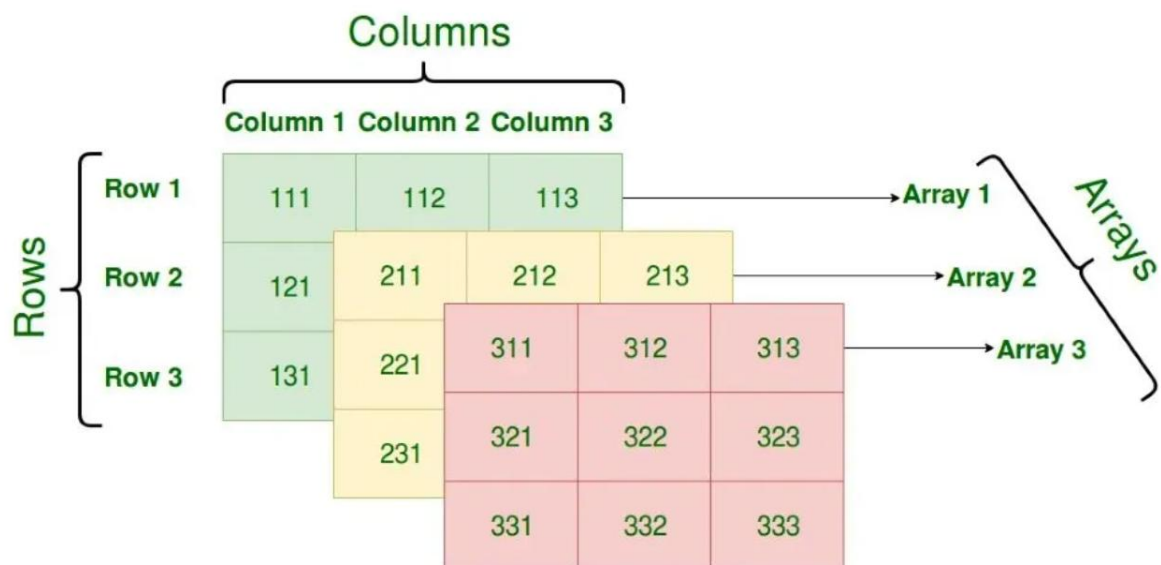
            System.out.println();
        }
    }
}

```

10 20 30
40 50 60 70 80
90 100

3D Array

```
int[][][] arr = new int[3][3][3];
```



```

public class Main {
    public static void main(String[] args) {
        int[][][] arr = new int[3][3][3];

        arr[0][0][0] = 111;
        arr[0][0][1] = 112;
        arr[0][0][2] = 113;
    }
}

```

```
arr[0][1][0] = 121;
arr[0][1][1] = 122;
arr[0][1][2] = 123;

arr[1][0][0] = 211;
arr[1][0][1] = 212;
arr[1][0][2] = 213;

arr[2][0][0] = 311;
arr[2][0][1] = 312;
arr[2][0][2] = 313;

System.out.println(arr[0][0][0]);
System.out.println(arr[1][0][1]);
System.out.println(arr[2][0][2]);
    }
}
```

111
212
313